

Beyond Spatial Accuracy: Diagnosing Persistent Orientation Failures in Vision–Language Models

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Abstract

Evaluating AI/ML systems responsibly requires practices for testing, stress-testing, auditing, comparing, and interpreting specific capabilities—not only aggregate benchmark scores. This paper proposes such a practice for spatial reasoning in vision–language models (VLMs): a diagnostic audit of object-relative orientation, which aggregate accuracy can hide. On Visual Spatial Reasoning (VSR), moving from a compact 2.2B model to a 7B VLM substantially improves overall, depth, horizontal, and containment accuracy, but orientation changes only modestly; a large 2026 sparse-MoE model reproduces the pattern (79.5% / 65.7% overall/orientation), a newer flagship (GPT-5.6 Sol) does not reproduce the same relative orientation-weakness pattern (73.0% orientation), and a third external model (Gemini Spark) yields only a degenerate evaluation (chance-level, TRUE-defaulting; its redo reused the orientation answers, App. A). Except for structured prompting, trained conditions on both controlled models and the sparse-MoE model remain in a narrow 62–66% orientation band. Prompting, LoRA variants, visual-side adaptation, probes, and object-centric decomposition do not reliably close the gap, and a label-sensitivity analysis (an automated audit followed by two blind human re-audits; humans agree well on the failure taxonomy, but exclusion masks remain annotator-dependent and automated auditors vary substantially in calibration) shows that the pattern persists after excluding examples flagged as ambiguous.

Accuracy also fails to capture relational coherence. LM-side adaptation improves consistency over zero-shot; hard-negative training raises facing consistency from 66.0% to 77.7% with facing accuracy unchanged at 68.9% (pooled gain over LM-only n.s.). Cross-dataset evaluation on SITE shows VSR-trained adaptation does not transfer cleanly: neutral overall, significantly harming official spatial-relationship reason-

ing (75.1%→71.6%, $p = 0.004$). A pre-specified orientation-vocabulary slice, frozen before SITE evaluation, is analyzed as exploratory evidence. The results point to an orientation bottleneck that is persistent in the controlled model families and the sparse-MoE model but not universal (GPT-5.6 Sol does not reproduce the same relative pattern; Gemini Spark degenerates), in which orientation is not robustly decodable by the tested readouts and relational decisions are not always propagated coherently, while task-specific spatial adaptation can transfer poorly. Each component of the audit—the diagnostic ladder, the label-sensitivity check, the accuracy–coherence separation, and the cross-dataset transfer protocol—is specified operationally so it can be applied to other capabilities, benchmarks, and model families.

1. Introduction

This paper contributes an evaluation practice as much as a set of model results. We develop a protocol for testing, stress-testing, auditing, comparing, and interpreting one capability across vision–language models (VLMs), and instantiate it for object-relative orientation: facing, facing away, parallel, and perpendicular. This capability matters for robotics, embodied agents, navigation, and scene understanding, and is easy to hide behind an aggregate score: identifying the objects, inferring the relation, and answering complementary formulations are separable skills.

Prior benchmarks establish that grounded spatial reasoning remains difficult, but they leave open how to diagnose a relation-specific failure. VSR introduced a controlled natural-image probe and reported particularly poor object-orientation performance [7]; What’sUp and GSR-Bench similarly expose failures on basic grounded relations [6, 9]; and broader evaluations continue to identify orientation as a difficult factor [15]. We therefore ask three narrower questions: does the

075 VSR orientation weakness persist in modern genera-
076 tive VLMs and at much larger scale; can prompting,
077 adaptation, or visual readouts remove it; and do accu-
078 racy, logical coherence, and cross-dataset transfer tell
079 the same story?

080 The novelty is diagnostic rather than a claim that
081 we discovered spatial difficulty. Our contribution is not
082 any individual diagnostic, but a falsification protocol
083 that holds the relation family and examples fixed while
084 testing whether an apparent capability failure survives
085 scaling and adaptation, label re-audit, representation
086 readout, logical reformulation, and cross-dataset trans-
087 fer. Prior work generally reports benchmark accuracy,
088 studies a single intervention, or proposes a consistency
089 objective; one controlled audit of the same family dis-
090 tinguishes a persistent relation-specific bottleneck from
091 aggregate underperformance, annotation artifacts, or
092 benchmark-specific specialization.

093 In our VSR experiments, the distinction between
094 relation families is striking. Moving from SmolVLM2-
095 2.2B-Instruct to the larger Qwen2-VL-7B-Instruct in-
096 creases zero-shot overall accuracy from 74.0% to 80.9%;
097 horizontal, depth, and containment rise by 6–15 pp.
098 Orientation, by contrast, moves only from 62.8% to
099 63.5%. A large 2026 sparse-MoE model (MiMo-V2.5,
100 310B total / 15B active parameters) reproduces the
101 same relation-specific pattern: 79.5% overall accuracy
102 yet 65.7% on orientation, again inside the 62–66%
103 band. Two further external zero-shot models bound
104 this result: GPT-5.6 Sol does not reproduce the rela-
105 tive weakness—73.0% orientation, above its own depth
106 (67.1%) and outside the band—while Gemini Spark’s
107 evaluation degenerates (51.5% overall; 98.6% TRUE
108 answers in its first pass, a redo that reused the 137 or-
109 ientation verdicts, App. A). The bottleneck is therefore
110 model-family-dependent rather than universal, and one
111 function of the protocol is precisely to detect such
112 differences instead of assuming a failure generalizes.
113 We then evaluate prompting, language-side adaptation,
114 targeted data construction, visual-side adaptation, rep-
115 resentation probing, and object-centric decomposition:
116 structured prompting significantly hurts; general and
117 targeted LoRA improve aggregate accuracy but do not
118 remove the persistent orientation weakness; hard neg-
119 atives improve targeted relational coherence without
120 a significant orientation-accuracy gain; and the tested
121 visual-side LoRA configurations do not outperform the
122 language-side control.

123 As a sensitivity analysis, removing examples flagged
124 by the automated audit raises orientation accuracy
125 (65.7%→78.5% for 7B General LoRA, 66.4%→76.6%
126 for hard-negative LoRA on the strictest 107-example
127 subset) but does not eliminate error (2B structured

shows no gain). The exclusion sets are annotator-
dependent: two blind human re-audits agree strongly
with each other on the failure taxonomy (Krippen-
dorff’s $\alpha=0.81$) while their clean/ambiguous flags agree
only moderately with the automated audit ($\kappa=0.36$ on
the 48 shared persistent-failure cases); details are in the
supplementary. Part of this gain is expected by con-
struction, because the removed examples were drawn
from persistent failures.

A second failure mode emerges from logical consis-
tency. The zero-shot 7B model is consistent on only
36.9% of complementary facing pairs. LM-side LoRA
substantially improves coherence, and hard negatives
raise facing consistency to 77.7% with facing accuracy
unchanged at 68.9%; the extra gain is not significant in
the pooled strict-family comparison, yet accuracy stays
fixed while consistency moves—the metrics capture dif-
ferent behavior.

Finally, we test cross-dataset transfer. The 7B
model is strong on SITE’s official spatial-relationship-
reasoning category (75.1% raw; 59.2% chance-adjusted
accuracy), yet the VSR-trained adapter does not
generalize cleanly: it is statistically neutral overall
(54.2%→53.8%, $p = 0.66$) and significantly degrades
the official category (75.1%→71.6%, $p = 0.004$). A
pre-specified, non-official orientation-vocabulary slice
is much weaker in the aggregate (48.3% raw; 24.0%
chance-adjusted) and shows no positive transfer, but—
as we detail below—its composition limits what it can
establish, and we therefore do not treat SITE as inde-
pendent confirmation of the VSR orientation construct.

Our contributions are four reusable components of
an evaluation protocol:

- A fine-grained diagnostic ladder. Across ten VSR
conditions, two differently sized models, and three
external zero-shot models, several spatial families
improve substantially while object-relative orienta-
tion remains unusually resistant in the controlled
families; one newer flagship does not reproduce
the same relative orientation-weakness pattern and
one external model’s evaluation degenerates under
the protocol, bounding rather than weakening the
claim. The ladder—scaling, prompting, language-
side adaptation, targeted data construction, visual-
side adaptation, representation probing, and ex-
plicit decomposition—is defined operationally so it
transfers to other benchmarks and relation families.
- A robustness and sensitivity audit. A label-
sensitivity analysis (automated labels with two inde-
pendent human re-audits), frozen/object-grounded
probes, visual-side adaptation, and object-centric
decomposition constrain purely annotation-based or
single-component explanations. The audit practice

181	(flagged-example exclusion sets, decodability checks	232
182	against majority baselines) applies to any bench-	233
183	mark whose labels or examples may be ambiguous.	234
184	• An accuracy-coherence separation protocol.	235
185	Complementary-relation tests on VSR's strict	236
186	relation pairs show that, for the tested conditions,	237
187	relational consistency can change substantially	238
188	while task accuracy remains fixed. The protocol	239
189	measures consistency against verdict-pattern base	240
190	rates, so it can be applied wherever complementary	241
191	formulations of the same relation exist.	242
192	• A cross-dataset transfer protocol. SITE shows	243
193	strong official spatial-relationship performance,	244
194	while VSR-trained LoRA transfers poorly and sig-	245
195	nificantly harms the official SITE category; the	246
196	orientation-vocabulary slice is analyzed as ex-	247
197	ploratory evidence. The protocol (pre-specified	248
198	subsets, chance-adjusted accuracy, paired transfer	249
199	tests) is model- and benchmark-agnostic and is re-	250
200	ported with negative as well as positive transfer.	
201	The protocol is deliberately modular: each compo-	
202	nent can be applied independently to other capabilities,	
203	benchmarks, and model families. The supplementary	
204	specifies the prompts, subsets, decision rules, and re-	
205	leased artifacts needed to reproduce the audit.	
206	2. Related Work	
207	Benchmarks for grounded spatial reasoning. VSR in-	
208	troduced a controlled natural-image benchmark with	
209	66 spatial relations and identified facing and parallel	
210	as difficult relations for the VLMs studied at the time [7].	
211	What'sUp controls object identity while varying spa-	
212	tial configuration and reports a substantial human-	
213	model gap [6]; GSR-Bench broadens grounded evalu-	
214	ation across models and scales [9]. SITE extends cov-	
215	erage to single-image, multi-image, and video inputs	
216	and organizes tasks by dimensions such as orientation	
217	and intrinsic/extrinsic relations [15]. SpatialScore and	
218	Mind the Gap further show that spatial competence	
219	remains fragmented [10, 16]. We use VSR not to re-	
220	establish that spatial reasoning is hard, but to turn its	
221	orientation signal into a controlled diagnostic target	
222	and use SITE to test transfer.	
223	Diagnosing and improving spatial failures. Several	
224	lines of work move beyond aggregate accuracy.	
225	AdaptVis relates successful VSR reasoning to attention	
226	on relevant object locations [2], while CREG shows that	
227	attribution evidence in Qwen2-VL can reflect image lay-	
228	out rather than the queried relation [11]. When More Is	
229	Less finds that spatial cues, commonsense information,	
230	and reasoning instructions can help or hurt depend-	
231	ing on the model and setup [1]. Other work probes	
	visual-concept decodability [13], separates perception	
	from reasoning in spatial planning [17], or supplies ge-	
	ometric inputs for 3D spatial understanding [3]. These	
	motivate our combination of frozen/readout diagnos-	
	tics with adaptation and object-centric decomposition,	
	but none audits one relation family across accuracy,	
	coherence, label sensitivity, and transfer.	
	Taken together, prior work establishes the problem	
	and offers several useful diagnostic lenses, but leaves	
	a methodological gap: benchmark accuracy, represen-	
	tation access, logical consistency, annotation sensitiv-	
	ity, and transfer are usually reported separately. We	
	close that gap with a single controlled diagnostic lad-	
	der, with analysis choices and external-evaluation sub-	
	sets frozen where specified, applied to the same rela-	
	tion family. Our claim is consequently not to discover	
	spatial difficulty, but to make a persistent orientation	
	weakness measurable, falsifiable, and reusable as an	
	evaluation protocol.	
	Adaptation, specialization, and transfer. Parameter-	
	efficient adaptation such as LoRA [5] can improve a	
	benchmark without improving the underlying capa-	
	bility; we therefore compare general, targeted, hard-	
	negative, projector, and vision-plus-projector updates	
	and evaluate the strongest adapter on an untouched	
	benchmark. VSR adaptation is neutral overall on	
	SITE, does not improve the orientation subset, and	
	significantly degrades the official spatial-relationship	
	category—negative transfer treated as an evaluation	
	result rather than an omitted outcome.	
	Representation probing and logical consistency.	
	Probes can test whether a signal is accessible from	
	frozen representations, but probe capacity can itself	
	learn the task, requiring control baselines and cautious	
	interpretation [4]. Consistency-oriented objectives	
	such as SAGE use geometric and linguistic dualities	
	to train spatial reasoning [8]. We instead use probing	
	and complementary-pair consistency as independent	
	diagnostics: the former asks what the tested readouts	
	can decode, while the latter asks whether answers	
	to logically linked statements agree, so accuracy and	
	coherence can diverge as diagnostic axes.	
	3. Experimental Setup	
	3.1. Benchmarks	
	VSR. We use VSR [7] as the primary diagnostic	
	benchmark. The random-split test set contains 2,195	
	examples. The benchmark defines 64 relation labels;	
	61 appear in the evaluated test split. The orientation	
	family contains facing, facing away from, parallel to,	

281 and perpendicular to, totaling 137 test examples. All
282 main VSR results use the same held-out test set and
283 deterministic True/False parsing.

284 SITE. We use SITE [15] for cross-dataset transfer
285 evaluation only and do not train on SITE. The image
286 evaluation contains 2,591 examples: 1,368 single-image
287 and 1,223 multi-image examples. Our primary SITE
288 subset is the official spatial-relationship-reasoning cate-
289 gory ($n = 993$). We additionally report a transparently
290 defined, non-official keyword-derived orientation subset
291 ($n = 2,024$), analyzed as exploratory evidence rather
292 than as direct replication of the VSR construct. SITE
293 results use the benchmark’s native multiple-choice for-
294 mat and chance-adjusted accuracy (CAA) in addition
295 to raw accuracy.

296 3.2. Models and Conditions

297 Our compact model is SmolVLM2-2.2B-Instruct, and
298 our larger controlled-study model is Qwen2-VL-7B-
299 Instruct [14]. We retain Qwen2-VL-7B as the con-
300 trolled anchor because it supports the complete local
301 ladder—the same base model across prompting, LoRA,
302 visual-side adaptation, probing, and decomposition—
303 rather than mixing intervention effects with an uncon-
304 trolled architecture change. We evaluate ten canon-
305 ical VSR conditions: 2B zero-shot, 2B structured
306 prompting, 2B General LoRA, 2B Targeted LoRA,
307 7B zero-shot, 7B General LoRA, 7B Targeted LoRA,
308 7B Hard-Neg LoRA, 7B Projector LoRA, and 7B
309 Vision+Projector LoRA. LoRA [5] is used for all
310 parameter-efficient adaptation conditions. We addi-
311 tionally evaluate a large zero-shot sparse-MoE model,
312 MiMo-V2.5 [12] (310B total / 15B activated param-
313 eters, dedicated 729M-parameter vision encoder, 2026),
314 served through its public API with the identical
315 frozen prompt, thinking disabled, and greedy decoding
316 (App. A). Its role is external validation of the diagnosed
317 pattern; it is not fine-tuned. Two additional exter-
318 nal zero-shot models used the same frozen prompt, the
319 same parser, and the same invalid-counted-wrong pol-
320 icy: GPT-5.6 Sol (OpenAI, per-item evaluation, 2026-
321 08-12; App. A) and Gemini Spark (Google, per-item
322 run `run_gemini_spark_vsr2195_v2_001`). Gemini’s
323 first pass defaulted to TRUE on 98.6% of items; its
324 redo reused the 137 orientation answers (id-level veri-
325 fied), so its Table 1 row is a degenerate-evaluation find-
326 ing, not a capability estimate (App. A).

327 General LoRA uses a proportional sample across
328 relation families; Targeted LoRA concentrates train-
329 ing on weak orientation/depth/horizontal families with
330 rehearsal of the rest; Hard-Neg LoRA adds au-
331 dited complementary orientation examples, primarily

facing/facing-away inversions; vision-side conditions
adapt the multimodal projector or upper vision blocks
with it.

335 3.3. Metrics and Statistical Tests

336 For VSR we report accuracy overall and by spatial fam-
337 ily. For SITE we report raw accuracy and CAA. Paired
338 condition comparisons use McNemar tests when ap-
339 plicable. Clean-label orientation analysis reports the
340 full orientation set and progressively stricter audited
341 subsets. Logical consistency is evaluated by construct-
342 ing complementary statements for strict relation pairs
343 (left/right, front/behind, facing/facing-away) and test-
344 ing whether the model assigns opposite verdicts as re-
345 quired. Parallel/perpendicular is treated separately as
346 a soft complement because both relations can be false
347 for oblique objects.

4. Main Results

4.1. Orientation Remains Weak Across Two Differ- ently Sized VLMs

351 Table 1 shows the central pattern. Moving from the 2B
352 model to the 7B model lifts zero-shot overall accuracy
353 by 6.9 pp, horizontal by 14.6 pp, depth by 6.3 pp, and
354 containment by 5.9 pp, but orientation by only 0.7 pp.
355 General 7B LoRA further raises overall accuracy to
356 84.7%, yet orientation reaches only 65.7%. Across
357 trained conditions other than structured prompting,
358 orientation remains in a narrow 62–66% band. The
359 large sparse-MoE zero-shot model (MiMo-V2.5, evalu-
360 ated identically, App. A) falls in the same band: 79.5%
361 overall and 65.7% orientation, with the orientation fam-
362 ily again its weakest relation family (per-family table
363 in the supplementary). The relation-specific pattern is
364 therefore not an artifact of the 2B/7B scale range. Two
365 further external zero-shot models bound the pattern:
366 GPT-5.6 Sol does not reproduce the relative weakness
367 (73.0% vs. its own 67.1% depth; CI [65.0, 79.7] over-
368 laps the band), while Gemini Spark’s evaluation degen-
369 erates (51.5% overall; App. A). The MiMo-V2.5 row
370 is also robust to output-format parsing: orientation
371 is unchanged parseable-only (65.7%), and parseable-
372 only overall is 82.1% vs. 79.5% counting the 69 non-
373 conforming outputs wrong (App. G).

374 A paired bootstrap over the 2,195 shared ex-
375 amples (10,000 resamples) confirms the horizontal-
376 vs-orientation scaling contrast (+13.9 pp; 95% CI
377 [+2.9, +24.8]; positive in 99.3% of resamples); depth
378 (+5.5 pp) and containment (+5.1 pp) are directionally
379 positive but not individually significant (App. D).

380 Structured prompting is a negative result: overall
381 accuracy falls from 74.0% to 68.3% at 2B, while ori-

Table 1. Canonical VSR test accuracy (%). Orientation improves little with model size or adaptation compared with several other relation families.

Condition	Overall	Orientation	Depth	Horizontal	Containment	Topology/contact
2B zero-shot	74.0	62.8	68.9	70.2	83.4	80.5
2B structured	68.3	53.3	64.9	66.7	78.7	71.7
2B General LoRA	76.6	62.0	71.1	74.3	87.6	81.0
2B Targeted LoRA	76.5	64.2	70.8	75.1	87.6	81.4
7B zero-shot	80.9	63.5	75.2	84.8	89.3	80.3
7B General LoRA	84.7	65.7	82.3	87.4	92.9	84.4
7B Targeted LoRA	83.9	64.2	81.7	87.1	91.7	85.3
7B Hard-Neg LoRA	84.3	66.4	80.7	87.1	89.9	84.6
7B Projector LoRA	82.9	64.2	78.6	87.1	89.3	85.3
7B Vision+Projector LoRA	83.1	64.2	78.0	87.1	89.9	84.4
MiMo-V2.5 zero-shot	79.5	65.7	74.8	81.9	87.1	84.6
GPT-5.6 Sol zero-shot	75.4	73.0	67.1	76.9	82.8	77.8
Gemini Spark zero-shot [†]	51.5	57.7	55.0	53.2	50.3	47.5

[†]Gemini Spark’s evaluation is degenerate (98.6% TRUE in pass 1; redo reused the orientation answers, id-level verified); the row is a degenerate-evaluation finding, not a capability estimate (App. A).

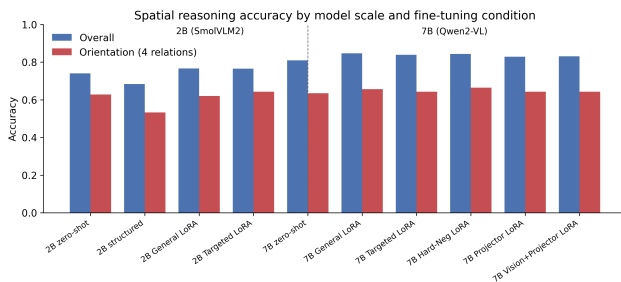


Figure 1. Overall versus orientation performance across the two VLMs and adaptation conditions.

entation falls from 62.8% to 53.3%. A paired exact McNemar test on the full test set confirms that the structured prompt is worse than the minimal baseline ($p \approx 2.2 \times 10^{-7}$). Consistent with recent VSR analyses where additional cues or reasoning instructions help or hurt depending on setup [1], this is specific to the tested 2B condition and prompt.

4.2. Clean-Label Sensitivity Analysis

An automated audit (LLM-generated labels; frozen v1) flagged ambiguous or questionable orientation examples; we repeat evaluation on progressively stricter subsets as a sensitivity analysis (full table in the supplementary). The best full-set orientation result is 66.4%; on the automated strictest 107-example subset, the best result rises to 78.5% for 7B General LoRA. Because auditing changes the sample, these are not matched comparisons across families; two blind human re-audits (binary flag on all 137; eight-class taxonomy on the 48 persistent failures) yield three distinct anal-

yses (App. F).

Taxonomy-derived strict masks. Each rater’s strict mask raises orientation accuracy without eliminating error—automated 78.5% ($n=107$), human A 77.6% ($n=107$), human B 78.8% ($n=104$)—and 2B structured prompting still shows no gain under any of the three masks (53.3%, 51.4%, 51.0%).

Binary clean/ambiguous masks. The two humans agree substantially on the 137-example binary flag ($\kappa=0.645$, 82.5% agreement) but only moderately with the automated audit ($\kappa=0.36$ on the 48 cases rated by both sources; the automated audit never flagged the remaining 89), and the taxonomy agrees only weakly with the automated labels ($\alpha=0.14$). Under the human binary masks the gains largely vanish (best 66.7%/66.2% vs. 66.4% full-set), tying gains to taxonomy-derived masks; audit reliability is itself model-dependent—GPT-5.6 Sol ($\kappa=0.34$) and Gemini Spark ($\kappa=0.03$, 126/137 clean) agree with humans only weakly—so LLM-generated ambiguity labels are sensitivity tools, not ground truth (App. F).

Consensus analysis. Under the consensus union mask ($n=62$), trained conditions again improve (7B General LoRA 77.4%) but 7B zero-shot does not (58.1%), and 2B structured’s no-gain pattern does not survive (69.3%), consistent with annotator dependence.

The robust conclusion is narrower: taxonomy-derived strict masks raise orientation scores but do not eliminate error across conditions; the audit is exploratory qualitative evidence, not a ruling on label noise (frozen v1 and both human tables in the supplementary).

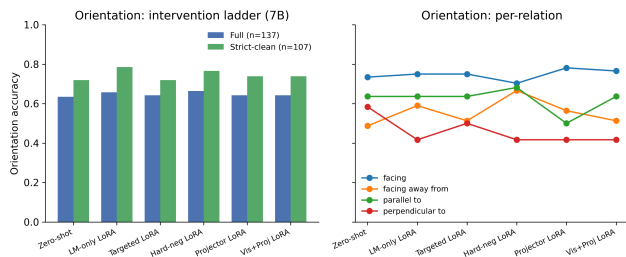


Figure 2. Orientation performance across 7B interventions, including clean-label analysis and per-relation behavior.

5. Why Does Orientation Remain Difficult?

5.1. Frozen Representation Probes

We probe frozen Qwen2-VL-7B-Instruct visual representations (no fine-tuning). Mean-pooled ViT and post-merger features support only weak relation decoding for facing versus facing-away, parallel versus perpendicular, and a four-way orientation task. Linear and MLP probes generally remain near majority baselines; preserving patch-level structure yields a suggestive point estimate for parallel/perpendicular (patch-vote 73.5% vs. 64.7% majority), but that test set is small ($n=34$) and the CI overlaps the majority baseline.

To reduce the confound that global features infer relations without identifying objects, we extract subject/reference regions and construct grounded features from subject, object, difference, and elementwise-product representations; object-box conditioning with the model’s own grounding predictions and mean-pooled region features does not recover robust orientation decodability. Following standard cautions about diagnostic classifiers [4], we phrase these results as decodability under the tested readouts, not as proof that the information is absent from the network. Probing decodability of visual concepts has precedent in driving-oriented VLMs [13] and attribution analyses of this model family [11]; ours targets VSR’s orientation family with pooled, patch-vote, and object-grounded readouts over the model’s exact conditioning features, paired with adaptation and consistency interventions. Detailed probe tables and plots are in the supplementary.

5.2. Visual-Side Adaptation and Object-Centric Decomposition

If the frozen representation were the sole limiting factor, adapting visual or projector components might recover orientation. For the trained checkpoints here it did not: projector and vision+projector LoRA both produced 64.2% orientation, below the 65.7% LM-only

control, with overall nominally lower (82.9%/83.1% vs. 84.7%; unadjusted $p=0.0043/0.0122$, not surviving blanket adjustment and not load-bearing, App. D).

We also test a box-based two-stage decomposition. Its four-way relation module classifies facing/facing-away/parallel/perpendicular against ground-truth relation labels at 44–46% cross-validation accuracy (49.9% majority baseline; uniform four-class chance 25%), so the tested relation classifier does not outperform the empirical majority baseline. Consistent with that, the pipeline’s verdicts match the control’s correctness on only 55.1–58.3% of the 127 boxed statements (per-relation agreement 25.0–69.4%, weakest for parallel); in the committed analysis (10 no-box statements scored wrong) they diverge from the control on 61–65 of 137, with the control correct and the pipeline wrong in 44–47 of them (App. D). These are agreement scores, not ground-truth accuracies: the committed script scored the pipeline against the control’s verdicts, so the decomposition is reported as an agreement diagnostic that fails to reproduce the model’s behavior, not as a direct accuracy comparison (ground-truth scoring via the released script, App. G). Two-dimensional box geometry is structurally insufficient for intrinsic facing direction, and the tested region representations do not close the gap.

5.3. Logical Consistency Reveals a Distinct Failure Mode

We construct complementary statements for strict relation families. A coherent model should return opposite truth values for left/right, front/behind, and facing/facing-away formulations of the same object pair. Complementary-pair consistency has prior literature, including consistency-oriented training [8]; our diagnostic below uses strict pairs with verdict-pattern and base-rate analysis. The zero-shot 7B model is consistent on only 58.0% of left/right pairs, 57.0% of front/behind pairs, and 36.9% of facing pairs. The large sparse-MoE model does not repair coherence: 42.7% facing-pair consistency vs. 36.9% (44.0% on the 100 parseable pairs, three excluded for non-conforming outputs), with the same False-defaulting pattern (0 both-True, 56.0% both-False; verdict table in the supplementary). Scale alone therefore does not resolve the coherence failure mode. The two external models diverge: GPT-5.6 Sol reaches 83.3% complementary facing verdicts (no both-True, $n=102$; 54.5% expected under independence), far above the controlled families, while Gemini Spark’s 60.2% with a zero flip-True rate and 100% complementary front/behind pairs are treated as degenerate, not capability estimates (App. A). The verdict patterns show a False-defaulting

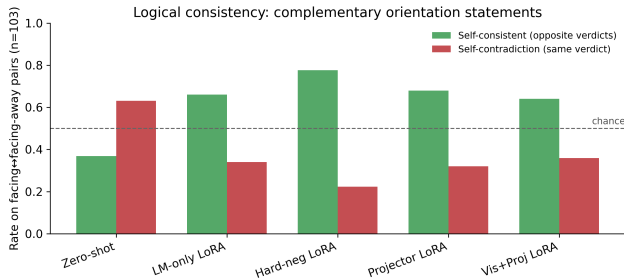


Figure 3. Self-consistency on complementary facing/facing-away relations across 7B conditions.

bias (both-False 63.1%, both-True 0%, above the 30.2% expected under verdict independence; verdict table in the supplementary).

LM-only LoRA raises facing consistency to 66.0% and front/behind to 70.7% (pooled strict-family significantly better than zero-shot, $p < 0.0001$); hard negatives raise facing consistency to 77.7% with facing accuracy identical at 68.9% (pooled gain over LM-only n.s., $p = 0.29$). The key result is separability: identical facing accuracy coexists with substantially different consistency. Training changes the failure mode rather than eliminating it: both-False facing verdicts drop from 63.1% to 9.7% while both-True contradictions rise from 0% to 24.3%, a disclosed trade-off.

6. Cross-Dataset Transfer to SITE

We evaluate the frozen 7B base model and the VSR-trained 7B General LoRA adapter on the same 2,591 SITE image examples using SITE’s native multiple-choice protocol [15], with no SITE training. Because the pre-specified decision rule resolved to branch 2, this adapter run is a post-protocol transfer check rather than a pre-specified step (protocol and outcome in the supplementary). The model is strong on the official spatial-relationship-reasoning category: 75.1% raw accuracy and 59.2% CAA. Applying the VSR-trained adapter does not transfer cleanly: it is statistically neutral overall (54.2%→53.8%, $p = 0.66$) and significantly degrades the official category (75.1%→71.6%, $p = 0.004$; 52 examples fixed vs. 87 broken). VSR gains therefore transfer poorly and can induce negative transfer; we do not equate benchmark-specific fine-tuning gains with general spatial-reasoning improvement.

The pre-specified orientation-vocabulary slice is much weaker in the aggregate (48.3% raw / 24.0% CAA vs. 75.1% / 59.2% official) with no positive transfer (48.3%→48.7%, $p = 0.70$). Because the slice is keyword-derived and heterogeneous (43% of its examples come from multi-view/cross-image reasoning, only

Table 2. SITE cross-dataset transfer (images). Entries are raw accuracy / CAA (%).

Subset	Zero-shot	VSR-LoRA	Δ
All images	54.2 / 31.1	53.8 / 30.5	-0.4
Official spatial-rel.	75.1 / 59.2	71.6 / 53.4	-3.5
Orient. heuristic	48.3 / 24.0	48.7 / 24.5	+0.4



Figure 4. SITE CAA for zero-shot and VSR-trained LoRA. The official spatial-relationship category degrades significantly under transfer.

24% from the spatial-relationship category), we treat it as exploratory evidence rather than replication of the VSR construct. A composition-confound analysis (supplementary §E) supports this caution: after controlling for official category, source dataset, modality, and option count, the orientation flag is not significantly associated with correctness (OR 0.84, 95% CI [0.60, 1.16], $p = 0.29$), though descriptively 7.5pp lower within the official category itself (71.3% vs. 78.8%, $n=488/505$). A post-hoc high-precision subset restricted to VSR-construct terms (facing, facing away, parallel, perpendicular; $n=177$) yields 44.1% raw (95% CI [37.0, 51.4]; CAA -2.3%) and is reported as exploratory only. We therefore do not treat SITE as independent confirmation of the VSR orientation bottleneck; the transfer results stand on the official category, where the model is strong and the adapter demonstrably does harm.

7. Discussion

Orientation is not spatial reasoning broadly. The strongest conclusion is deliberately narrow. The 7B model is strong on several VSR families and SITE’s official spatial-relationship category. The persistent weakness concerns the object-relative orientation family in VSR; on SITE, the orientation-vocabulary slice is analyzed as exploratory evidence rather than as independent confirmation. This framing is consistent with the original VSR observation [7] while extending it to modern generative VLMs and a broader intervention analysis. The weakness is also model-family-dependent: the

592 controlled families and the sparse-MoE model persist
593 in it, GPT-5.6 Sol does not reproduce it, and Gemini
594 Spark’s evaluation degenerates under the protocol; a
595 diagnostic that only found the failure would be unin-
596 formative, so this distinguishability is what makes the
597 protocol falsifiable.

598 Decodability and coherence both matter, as diagnos-
599 tics. The probes show that orientation is not robustly
600 decodable under the tested frozen-feature readouts,
601 while consistency experiments expose large logical fail-
602 ures even for relations answered above chance. Neither
603 alone identifies a causal mechanism; together they sup-
604 port a mixed account in which orientation information
605 is hard to access under the tested readouts and rela-
606 tional decisions are not always propagated coherently.

607 Task adaptation can specialize rather than generalize.
608 General LoRA improves VSR overall accuracy substan-
609 tially, but its SITE transfer is neutral overall and signif-
610 icantly negative on the official subset. Hard negatives
611 improve a targeted coherence behavior with relation-
612 specific tradeoffs. Benchmark-specific fine-tuning gains
613 should therefore not be equated with general spatial-
614 reasoning improvement.

615 8. Limitations

616 Statistical power is limited at the scale of the orien-
617 tation diagnostic. The VSR orientation family con-
618 tains only 137 test examples, and the smallest per-
619 relation subset, perpendicular, has $n=12$; perpendic-
620 ular results are descriptive only. The Wilson 95% con-
621 fidence intervals on the family-level point estimates are
622 correspondingly wide—62.8% [54, 70], 63.5% [55, 71],
623 65.7% [57, 73], and 66.4% [58, 74] for 2B zero-shot, 7B
624 zero-shot, 7B General LoRA, and 7B Hard-Neg LoRA
625 ($n=137$ each)—so the trained conditions’ 62–66% band
626 should be read as a set of statistically overlapping point
627 estimates rather than a fine-grained ranking. Several
628 smaller point estimates have confidence intervals that
629 include the majority baseline or chance: the patch-
630 vote parallel/perpendicular probe (73.5%, $n=34$; 95%
631 CI [57, 85] vs. the 64.7% majority), the perpendic-
632 ular subset for the trained 7B conditions (41.7%, $n=12$;
633 [19, 68] vs. 50%), and the 7B zero-shot facing-away sub-
634 set (48.7%, $n=39$; [34, 64]). Paired McNemar tests on
635 orientation rest on small discordant-pair counts: the
636 non-significant projector and vision+projector compar-
637 isons ($p=0.86$, $p=0.85$) indicate no detectable effect,
638 not equivalence; conclusions rely on the full-set paired
639 tests and the Holm-adjusted battery (supplementary).

640 The clean-label audit’s exclusion sets originate from
641 an automated LLM audit: two independent human

642 re-audits reached only moderate per-item agreement
643 with it ($\kappa=0.36$ on the clean/ambiguous flag; Krip-
644 pendorff’s $\alpha=0.14$ on the eight-class taxonomy) while
645 agreeing strongly with each other on the failure taxon-
646 omy ($\alpha=0.81$); both reproduced the qualitative conclu-
647 sions under their own taxonomy-derived strict masks,
648 so the sets are annotator-dependent (tables versioned:
649 frozen v1 plus both human re-audits). The SITE
650 orientation slice is keyword-derived; a composition-
651 confound analysis shows its aggregate score is largely
652 explained by task composition, so SITE is not inde-
653 pendent confirmation of the VSR bottleneck. Our
654 probing methods test accessible decodability under spe-
655 cific linear/MLP and pooling choices; failure to decode
656 does not prove information is absent. Visual-side ex-
657 periments use LoRA rather than full retraining; the
658 three model families form a cross-model comparison,
659 not a controlled scale experiment. All trained con-
660 ditions were single-checkpoint runs: paired McNemar
661 tests quantify disagreement on the fixed test exam-
662 ples, not training-run variability. A completed multi-
663 seed extension (3 seeds for 7B General LoRA, 2 each
664 for Targeted and Hard-Neg; identical recipe, seed-only
665 change) shows overall accuracy is stable across seeds
666 (± 0.1 – 0.4 pp SD per condition) while orientation accu-
667 racy varies ± 0.5 – 3.1 pp; every between-condition orien-
668 tation delta falls inside the seed-to-seed SD (App. D).
669 The MiMo-V2.5 evaluation is served through a public
670 API: 3.1% of outputs did not conform to the one-word
671 format and are counted wrong (never guessed), and
672 294 contained chain-of-thought text alongside the ver-
673 dict; both are disclosed in the committed predictions
674 (App. A). The MiMo-V2.5 row was evaluated on 2026-
675 08-12 through the OpenCode Go gateway; API-served
676 models can change without notice, so this row’s repro-
677 ducibility depends on the serving snapshot (recorded
678 in results/mimo/). The two additional external rows
679 (App. A) used the identical frozen prompt, parser,
680 and invalid-counted-wrong policy via per-item requests
681 on 2026-08-12, with raw outputs and per-item records
682 committed; API-served models can change without no-
683 tice, so these rows share the same serving-snapshot
684 caveat. Gemini Spark’s evaluation is degenerate (98.6%
685 TRUE in pass 1; redo reused 1,315/2,195 answers in-
686 cluding all 137 orientation verdicts, id-level verified),
687 so its row and coherence numbers are a documented
688 degenerate-evaluation outcome, not a capability esti-
689 mate. The SITE evaluation is image-only; SITE video
690 reasoning remains future work. Finally, several inter-
691 ventions are evaluated as negative results; lack of im-
692 provement holds for the tested configurations, not as
693 impossibility claims.

9. Conclusion

Aggregate accuracy hides an orientation bottleneck: adaptation improves several families while leaving orientation resistant, a sparse-MoE model reproduces the pattern, a newer flagship does not reproduce the same relative pattern, and one external model degenerates under the protocol. Auditing, probing, and decomposition constrain simple explanations; accuracy and coherence are distinct axes that remain weak in the large model; SITE shows VSR-specific adaptation can negatively transfer. Evaluation should report relation-specific performance, transfer, and logical coherence rather than aggregate accuracy.

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